



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

|               |
|---------------|
| JEE(Advanced) |
| UNIT TEST     |
| 12-06-2022    |

## JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS]

Time : 3 Hours

Maximum Marks : 216

### READ THE INSTRUCTIONS CAREFULLY

#### GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **20** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

#### OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

#### DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take  **$g = 10 \text{ m/s}^2$**  unless otherwise stated.

#### QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

**SOME USEFUL CONSTANTS**

**Atomic No.** : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

**Atomic masses :** H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- |                                    |                                                              |
|------------------------------------|--------------------------------------------------------------|
| • Boltzmann constant               | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$                   |
| • Coulomb's law constant           | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$                   |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$  |
| • Speed of light in vacuum         | $c = 3 \times 10^8 \text{ ms}^{-1}$                          |
| • Stefan-Boltzmann constant        | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$                        |
| • Permeability of vacuum           | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$                |
| • Permittivity of vacuum           | $\epsilon_0 = \frac{1}{\mu_0 c^2}$                           |
| • Planck constant                  | $h = 6.63 \times 10^{-34} \text{ J-s}$                       |

Space for Rough Work

**HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS**

**BEWARE OF NEGATIVE MARKING**

**PART-1 : PHYSICS**

**SECTION-I (i) : (Maximum Marks: 32)**

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.

*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. We assume a model of singly ionized Helium atom where nucleus is a uniformly charged sphere of radius  $a_0$  (Bohr radius) and charge  $+2e$ . The electron moves in a circular path about the centre of the nucleus. Assume Bohr quantization rule to be true. Radius of orbit can be smaller or greater than  $a_0$ .

(A) Radius of orbit for ground state is  $\sqrt{\frac{h}{e} \sqrt{\frac{\epsilon_0 a_0^3}{2\pi m}}}$

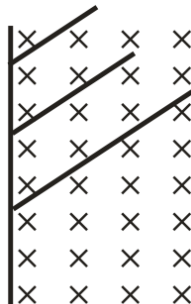
(B) The orbit for 1<sup>st</sup> excited state is outside the nucleus.

(C) The orbit for ground state is on the surface of the nucleus

(D) Speed of electron in ground state is  $\sqrt{\frac{he}{\sqrt{2\pi m \epsilon_0 a_0^3}}}$

**Space for Rough Work**

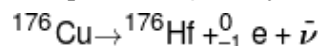
2. Photon of wavelength 310 nm are incident on calcium plate ( $\phi = 3.1$  eV). A uniform magnetic field of 1.5 T is parallel to the plate as shown. Take mass of electron as  $9 \times 10^{-31}$  kg. Choose the CORRECT statement (s). Neglect electrostatic interaction or gravity.



- (A) The speed of emitted photoelectron can be  $10^6$  m/s.
- (B) The emitted electron can move a maximum distance of 5  $\mu\text{m}$  from the plate.
- (C) The emitted electron can move a maximum distance of 3  $\mu\text{m}$  from the plate.
- (D) The speed of emitted photoelectron can be  $2 \times 10^5$  m/s.
3. Equal masses of two samples of charcoal A & B are burnt separately and resulting  $\text{CO}_2$  is collected in 2 separate vessels. The radioactivity of  $^{14}\text{C}$  is measured for both gas samples. The gas from charcoal A gives 2800 counts per week and gas from charcoal B gives 1400 counts per week. Half life of  $^{14}\text{C}$  is 5700 years:
- (A) The activity of carbon before and after burning is the same.
- (B) Charcoal B is older than charcoal A.
- (C) Charcoal A is older than charcoal B.
- (D) The difference in their lives is 5700 years.

Space for Rough Work

4. In a particular  $\beta$  decay of



Assume antineutrino to be like a photon

$$M_{\text{Cu}} = 175.942694 \text{ amu}$$

$$M_{\text{Hf}} = 175.941420 \text{ amu}$$

$$1 \text{ amu} = 931 \text{ MeV}$$

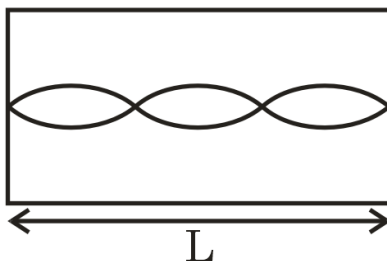
$$m_e = 0.51 \text{ MeV}.$$

- (A) The maximum possible kinetic energy of electron emitted is approx 1.18 eV.
- (B) The maximum possible kinetic energy of emitted electron is approx 0.16 MeV.
- (C) Minimum possible wavelength of antineutrino is approx 1.05 pm.
- (D) Minimum possible wavelength of antineutrino is approx 1.84 pm.
5. Which of the following statements are true about strong nuclear forces?
- (A) They don't act on electrons
- (B) They are spin dependent
- (C) They cause  $\beta^-$  decay
- (D) They are responsible for parallel alignment of atomic dipoles in ferromagnetic substance (domain formation).

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**Space for Rough Work**

6. Consider a particle of mass  $m$  forming standing matter wave in a one dimensional box of length  $L$ . Which of the following statement(s) are correct?

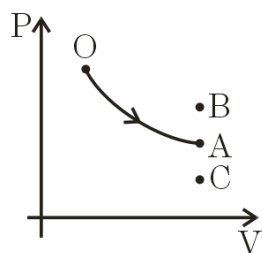


- (A) The particle can't be at rest.
- (B) The minimum energy of particle is  $\frac{h^2}{2mL^2}$
- (C) The particle exerts an average force of  $\frac{h^2}{8mL^3}$  on the walls in ground state.
- (D) The average force exerted by the particle on the walls is proportional to quantum number  $n$ .

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Space for Rough Work

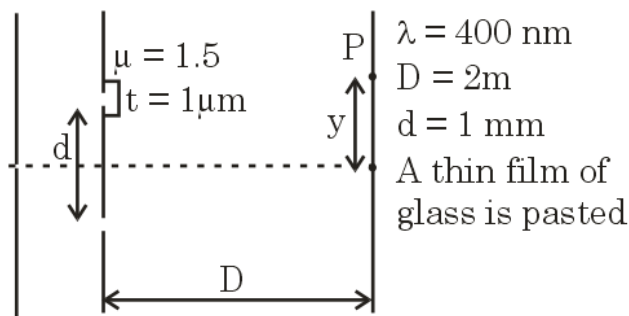
7. The arrow OA in the PV diagram represents a reversible adiabatic expansion of an ideal gas. The gas may undergo:



- (i) irreversible adiabatic free expansion  
(ii) reversible isothermal expansion.

starting from the same initial point O. Match the possible final state of gas with process. The graph is not drawn to scale.

- (A) (i)  $\rightarrow$  B                      (B) (i)  $\rightarrow$  A                      (C) (ii)  $\rightarrow$  B                      (D) (ii)  $\rightarrow$  (C)
8. Consider a YDSE apparatus using a glass slab in front of upper slit. We want to find the phase difference between wave coming from upper slit & lower slit at point P. Match  $\Delta\phi$  with y.



- (A)  $y = 1\text{ mm}$ ,  $\Delta\phi = 0$                       (B)  $y = 0.5\text{ mm}$ ,  $\Delta\phi = \frac{5\pi}{4}$
- (C)  $y = \frac{2}{3}\text{ mm}$ ;  $\Delta\phi = \frac{5\pi}{6}$                       (D)  $y = \frac{1}{3}\text{ mm}$ ,  $\Delta\phi = \frac{5\pi}{3}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:  
**Full Marks** : +4 If only (all) the correct option(s) is (are) chosen.  
**Partial Marks** : +3 If all the four options are correct but **ONLY** three options are chosen.  
**Partial Marks** : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.  
**Partial Marks** : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.  
**Zero Marks** : 0 If none of the options is chosen (i.e. the question is unanswered).  
**Negative Marks** : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

Paragraph for Questions 9 and 10

An electromagnetic wave given by  $\vec{E} = 10\hat{j} \sin(2\pi(x - ct))$  is travelling in empty space where there is no gravity. We have a bead carrying charge of  $200 \mu\text{C}$  and mass  $1 \text{ gm}$  threaded on a frictionless fixed straight rod. As the wave passes, the bead will experience a force due to electric and/or magnetic field of the wave. In addition, it may experience a force due to the rod. All quantities in expression for electric field and in questions below are in SI units.

9. If the bead is at rest, the normal force exerted by the rod at  $x = \frac{1}{4} \text{ m}$ ,  $t = 0$  is:
- (A)  $2 \times 10^{-3} \hat{j}$  if rod is oriented along x axis.  
 (B)  $2 \times 10^{-3} \hat{k}$  if rod is oriented along y axis.  
 (C)  $\vec{0}$  if rod is oriented along y axis.  
 (D)  $-2 \times 10^{-3} \hat{j}$  if rod is oriented along z axis.
10. If bead is moving with a velocity of  $10^7 \text{ m/s}$  along the rod, it's acceleration at  $x = \frac{1}{4} \text{ m}$ ,  $t = 0$  is:
- (A)  $2 \text{ m/s}^2$  if rod is oriented along y axis.      (B)  $\frac{1}{15} \text{ m/s}^2$  if rod is oriented along x axis.  
 (C)  $\frac{1}{15} \text{ m/s}^2$  if rod is oriented along z axis.      (D)  $0 \text{ m/s}^2$  if rod is oriented along z axis.

**Paragraph for Questions 11 and 12**

Into a vacuum vessel with a capacity of 1.5 ℓ with rigid and non-heat-conducting walls,  $m = 1.1$  g of  $T_2O$  liquid heavy water was injected at temperature of  $0^\circ C$  and then we sealed the vessel. The half-life of tritium nuclei  $t = 12$  years. In decay of one mole of tritium nuclei 1.8 GJ energy is released. The molar heat capacity of heavy water is the same as the molar heat capacity of ordinary water, i.e.

$75.6 \text{ J/ (mol} \times \text{K)}$ . The heat of vaporization is also such same as for ordinary water, i.e.  $L = 40 \text{ kJ / mol}$ .

Neglect the heat capacity of the vessel.  $N = 6 \times 10^{23}$  / mole,  $R = 25/3 \text{ J/mol K}$ , neglect vibrational degrees of freedom for gas. Take  $\ln 2 = 0.7$

11. Choose the correct statement(s):
- (A) Heat is generated at a constant rate.
  - (B) The rate of generation of heat decreases with time.
  - (C) The initial rate of heat generation is approx. 0.16 W.
  - (D) The initial rate of heat generation is approx 0.33 W.
12. If  $W = 10.0 \pm 0.1 \text{ N}$ ,  $\ell = 1.00 \pm 0.01 \text{ m}$ ,  $b = 10.0 \pm 0.1 \text{ cm}$ ,  $d = 1.00 \pm 0.01 \text{ cm}$ ,  $\delta = 1.00 \pm 0.01 \text{ mm}$ ,
- (A)  $Y = 2.5 \times 10^{10} \text{ N/m}^2$
  - (B)  $\Delta Y = 3.75 \times 10^9 \text{ N/m}^2$
  - (C)  $Y = 5 \times 10^{10} \text{ N/m}^2$
  - (D)  $\Delta Y = 2.25 \times 10^9 \text{ N/m}^2$

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**Space for Rough Work**

**SECTION-II : (Maximum Marks: 24)**

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +3 If ONLY the correct numerical value is entered.

*Zero Marks* : 0 In all other cases.

- 
1. Plane waves of blue light  $\lambda = 420$  nm fall on a single slit and then pass through a lens of focal length 80 cm (as in fraunhofer diffraction). The central maxima of diffraction pattern on the screen has width of 1.05 mm. What is the width of the single slit (in mm)?
  2. A thin film of thickness  $t$  and refractive index 1.5 is held perpendicular to light of wavelength 480 nm. What is the minimum non zero thickness (in  $10^{-8}$  m) of film so that reflected light shows destructive interference?
  3. A gas of identical hydrogen like atoms has some atoms in the lowest energy level A and some atoms in a particular upper level B and there are no atoms in any other energy level. The atoms of the gas make transition to a higher energy level by absorbing monochromatic light of photon energy 7.56 eV. Subsequently, the atoms emit radiation of only six different photon energies. Some of emitted photons have energy 7.56 eV, some have energy more and some have less than 7.56 eV. Calculate ionisation energy for gas atoms (in eV).
  4. A radioactive sample emits  $n$   $\beta$ -particles in 2 second. In next 2 sec it emits  $0.75n$   $\beta$ -particle, what is the mean life (in second) of the sample? [Take  $\ln 2 = 0.7$ ,  $\ln 3 = 1.1$ ]
- 

**Space for Rough Work**

5. Two large black parallel plates at temperatures  $127^{\circ}\text{C}$  and  $27^{\circ}\text{C}$  are held at a distance of  $1\text{m}$  in vacuum. If space in between is completely filled with material of conductivity  $2.1\text{ W/mK}$ , what is the ratio of new rate of heat flow to the original rate of heat flow? Take  $\sigma = 6 \times 10^{-8}\text{ W/m}^2\text{K}^4$ .



6. A steel tape measures the length of a copper rod as  $90.0\text{ cm}$  when both are at  $10^{\circ}\text{C}$ , which is the graduation temperature of the tape. What would the tape read (in  $\text{cm}$ ) for the length of the rod when both are at  $30^{\circ}\text{C}$ ? Given that coefficient of linear expansion of steel is  $1.2 \times 10^{-5}\text{ }^{\circ}\text{C}^{-1}$  and that of copper is  $1.7 \times 10^{-5}\text{ }^{\circ}\text{C}^{-1}$ .
7. A column of mercury is in the middle of a horizontal tube of uniform bore closed at both ends. The pressure of the enclosed air column on either side of mercury being  $75\text{cm}$  (of mercury). When the tube is placed vertical the length of the air column below mercury column is  $3/5$ th of that above it. Find the length (in  $\text{cm}$ ) of the mercury column. ( $g = 10\text{ m/s}^2$ )
8. A beam of photons with wavelength of  $310\text{ nm}$  is incident on a metal for which threshold wavelength is  $620\text{ nm}$ . Consider a photon which is absorbed by an electron. The electron then collides with the metal lattice due to which it loses  $0.5\text{ eV}$ . What is the de-broglie wavelength (in  $\text{\AA}$ ) of this electron after coming out?

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**Space for Rough Work**

**PART-2 : CHEMISTRY**  
**SECTION-I (i) : (Maximum Marks: 32)**

- This section contains **EIGHT (08)** questions.
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- 
1. Three complex compounds I, II and III are  $[M(H_2O)_6]^{3+}$ ,  $[M(en)_3]^{3+}$ ,  $[M(CN)_6]^{3-}$  respectively. Complex I and III have green and red colour respectively then identify the correct matching colour for complex II.

(A) Blue

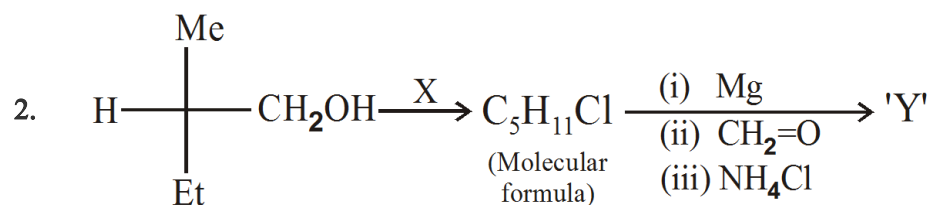
(B) Orange

(C) Yellow

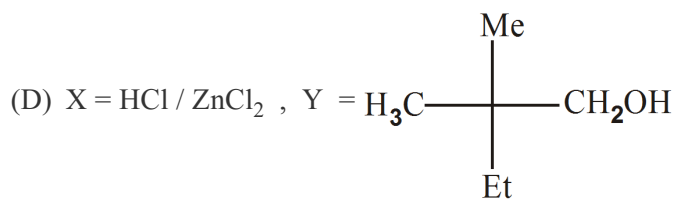
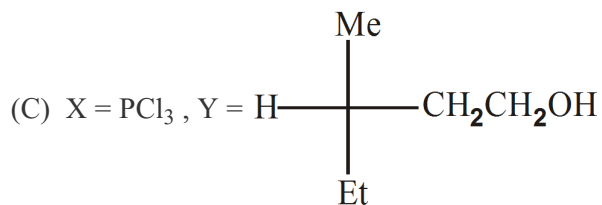
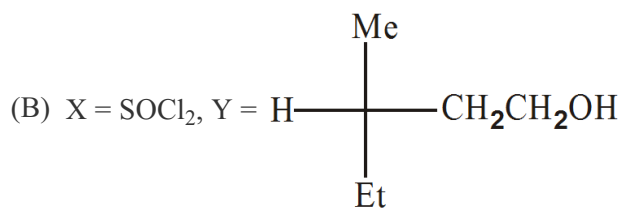
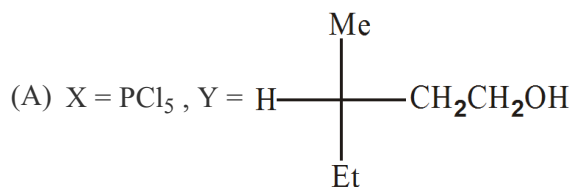
(D) Violet

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**Space for Rough Work**

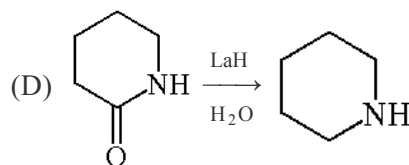
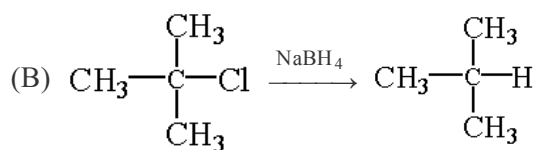
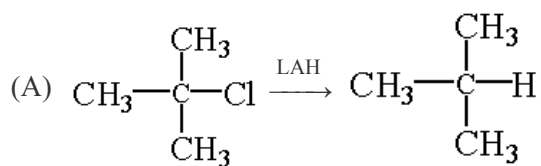


Correct pair of X & Y respectively is/are :



Space for Rough Work

3. Identify reaction correctly matched, with their major product (or statement written with them) :



4. Colour of iodine solution is disappeared by shaking it with aqueous solution of :-



5. For non-stoichiometric cuprous oxide,  $\text{Cu}_{1.80}\text{O}_{1.00}$ , select the correct information(s). (Cu = 63.5) :

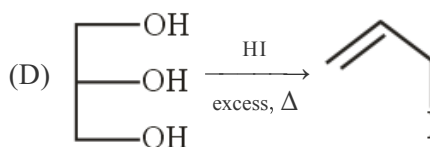
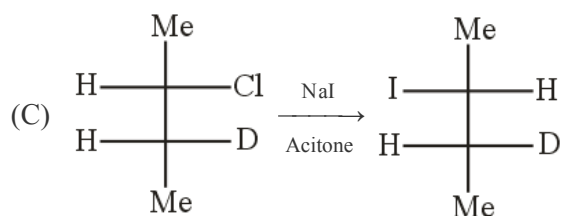
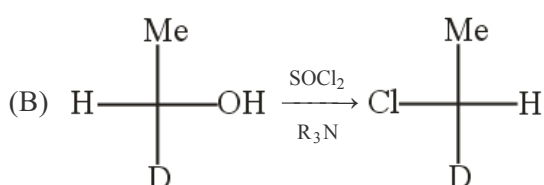
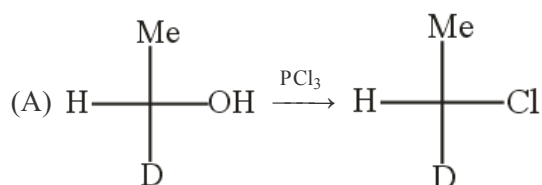
- (A) It behaves as p-type semiconductor  
 (B) % of total copper as Cu(I) is 11.11%  
 (C) It has metal deficiency defect  
 (D) Density of solid is 20% less than the ideal  $\text{Cu}_2\text{O}$  crystal

Space for Rough Work

6. Liquids 'A' and 'B' form an ideal solution. At  $25^{\circ}\text{C}$ ,  $P_A^{\circ} = 25 \text{ cm Hg}$  and  $P_B^{\circ} = 75 \text{ cm Hg}$ . If 5.0 moles of liquid 'A' is mixed with 20.0 moles of liquid 'B' and the pressure over the system is maintained at 50 cm Hg, the correct information(s) regarding the final composition of system is/are :

- (A) Vapour contains 5.0 moles of 'A'.  
 (B) Liquid solution contains 50% of 'A', by moles.  
 (C) Vapours contains 75% of 'B', by moles  
 (D) 50% of the total moles of liquid taken will vaporise.

7. Select reaction(s) with correct major product :



8. Which of the following metal sulphides dissolved in hot and dil.  $\text{HNO}_3$  ?

- (A)  $\text{HgS}$  (B)  $\text{ZnS}$  (C)  $\text{CuS}$  (D)  $\text{PbS}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

Paragraph for Questions 9 and 10

An octahedral homoleptic high spin complex of '3d' series central metal ion having  $3d^n$  electronic configuration have  $t_{2g}^x e_g^2$  arrangement of electrons.

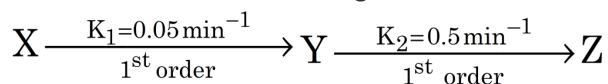
Answer the following based on above paragraph.

9. Choose the correct pair of central atom/ion and ligand for the above mentioned complex.
- |                        |                        |
|------------------------|------------------------|
| (A) $Fe^{2+}$ , $NH_3$ | (B) $Co^{3+}$ , $NH_3$ |
| (C) $Fe^{3+}$ , $F^-$  | (D) $Ni^{2+}$ , $H_2O$ |
10. For the maximum possible value of 'x' the value of spin only magnetic moment will be.
- |                      |                       |                       |                       |
|----------------------|-----------------------|-----------------------|-----------------------|
| (A) $\sqrt{8}$ B. M. | (B) $\sqrt{15}$ B. M. | (C) $\sqrt{24}$ B. M. | (D) $\sqrt{35}$ B. M. |
|----------------------|-----------------------|-----------------------|-----------------------|

Space for Rough Work

**Paragraph for Questions 11 and 12**

A substance 'X' converts into 'Z' through some intermediate, 'Y'.



$t = 0$      $0.50\text{M}$                        $0$                        $0$

11. The correct concentration(s) of 'X', 'Y' and 'Z' at  $t = 20$  min is ( $e^{-1} = 0.36$ ,  $e^{-10} = 4.5 \times 10^{-5}$ ) :
- (A)  $[X] = 0.18 \text{ M}$
- (B)  $[Y] \approx 0.02 \text{ M}$
- (C)  $[Z] \approx 0.30 \text{ M}$
- (D)  $[X] + [Y] + [Z] = 0.50 \text{ M}$
12. The correct information(s) is/are ( $\ln 2 = 0.7$ ,  $\ln = 1.6$ ) :
- (A) The concentration of 'Y' becomes maximum at about 5.11 min.
- (B) The rate of formation of 'Z' becomes maximum at about 5.11 min.
- (C) The rate of disappearance of 'X' becomes equal to the rate of appearance of 'y' at about 5.11 min.
- (D) Concentration of 'X' becomes 0.25 M at about 5.11 min.

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**Space for Rough Work**

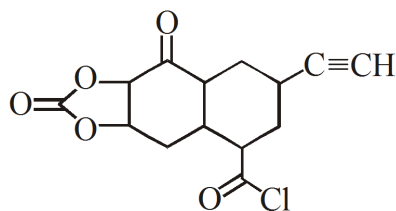
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +3 If ONLY the correct numerical value is entered.

*Zero Marks* : 0 In all other cases.

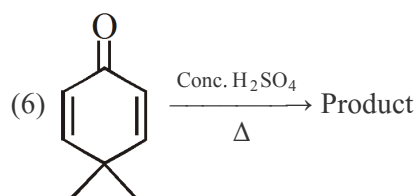
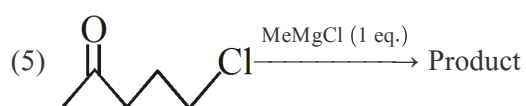
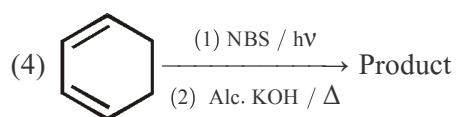
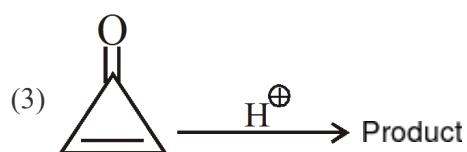
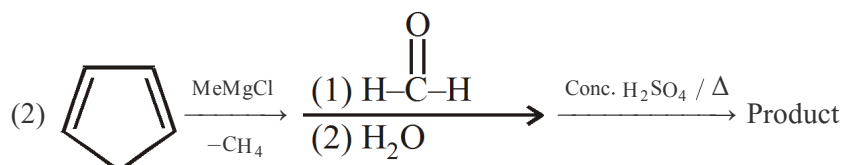
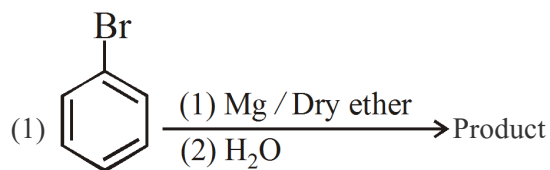
1. An octahedral complex  $\text{Ma}_2\text{b}_p\text{C}_q$  have 'X' number of cis isomer w.r.t. 'a' ligand and 'Y' number of trans isomers w.r.t. 'a' ligand. Find the value of (X+Y)  
(Consider the maximum possible value of X and Y)  
a, b, c are monodentate achiral ligands while p and q are non-zero positive values.
2. Total number of grignard reagent consumed by compound 'X' :



(X)

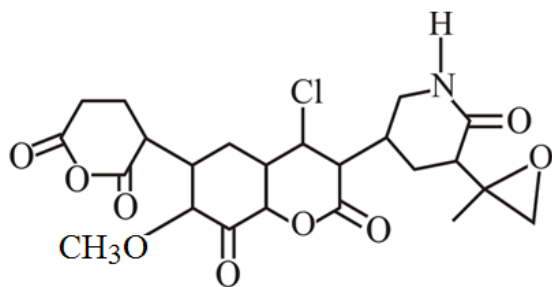
Space for Rough Work

3. Total number of reactions which give aromatic product as a major product ?



Space for Rough Work

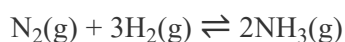
4. How many hydroxyl group present by reduction by  $\text{LiAlH}_4$  of given compound



5. Total number of following reagents, which can be used to distinguish (with visible change) between  $\text{SO}_2$  and  $\text{CO}_2$  gases :-

$\text{FeCl}_3$ -solution,  $\text{Br}_2$ -water,  $\text{Cl}_2$ -water,  $\text{H}_2\text{O}_2$ ,  $(\text{KMnO}_4 + \text{H}_2\text{SO}_4)$ , (starch- $\text{IO}_3^-$ ),  $(\text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4)$ ,  $\text{Ca}(\text{OH})_2$

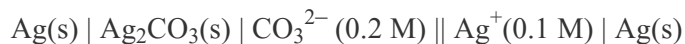
6.  $\text{N}_2$  and  $\text{H}_2$  gases are mixed in 1 : 3 mole ratio. At equilibrium at 100 atm, the equilibrium mixture is found to contain 20 mole percent  $\text{NH}_3$ . For the reaction :



$K_p$  is  $\left(\frac{x}{216}\right) \text{ atm}^{-2}$ . The value of 'x' is.

7. 2 moles of an ideal gas undergoes reversible isothermal expansion from  $5 \text{ dm}^3$  to  $20 \text{ dm}^3$  at 500 K. The amount of heat absorbed by the gas (in Kcal) is ( $\ln 2 = 0.693$ ).

8. For the Galvanic Cell :



$E_{\text{cell}} = 0.18 \text{ V}$  at 298 K.  $K_{\text{sp}}$  of  $\text{Ag}_2\text{CO}_3$  (in the order of  $10^{-9} \text{ M}^3$ ) is.

$$\left[ \frac{2.303 \text{ RT}}{F} = 0.06 \text{ V}, E_{\text{Ag}^+/\text{Ag}}^\circ = 0.80 \text{ V} \right]$$

Space for Rough Work

**PART-3 : MATHEMATICS**  
**SECTION-I (i) : (Maximum Marks: 32)**

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.

*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let  $x = \sum_{i=1}^{10} \frac{1}{10\sqrt{3}} \frac{1}{\left[1 + \left(\frac{i}{10\sqrt{3}}\right)^2\right]}$  and  $y = \sum_{i=0}^9 \frac{1}{10\sqrt{3}} \frac{1}{\left[1 + \left(\frac{i}{10\sqrt{3}}\right)^2\right]}$ , then -

(A)  $x < \frac{\pi}{6}$

(B)  $y > \frac{\pi}{6}$

(C)  $x + y < \frac{\pi}{3}$

(D)  $x + y > \frac{\pi}{3}$

**Space for Rough Work**

2. Let A denotes the area bounded by  $y = \ln^4 x$  and lines  $y = 0$ ,  $x = 1$  and  $x = e^4$ , then -
- (A)  $A < 256e^4 - 175e^3 - 81$  (B)  $A < 256e^4 - 256$
- (C)  $A > 256(e^4 - 1)$  (D)  $A > \frac{337}{2}e^3(e - 1)$
3. If  $\int \frac{3z^3 - 8z + 5}{\sqrt{z^2 - 4z - 7}} dz = (z^2 + az + 36) \sqrt{z^2 - 4z - 7} + b \ln |z - 2 + \sqrt{z^2 - 4z - 7}| + C$ , where  $a, b \in \mathbb{I}$  and C is integration constant, then -
- (A)  $a > b$  (B)  $a < b$
- (C)  $a + b = 117$  (D) exactly one out of a or b is a prime number
4. Let  $f(x) = \cos^{-1} \left( \ln \left( \frac{x+1}{x} \right) \right)$ , then which of the following statement(s) is/are correct
- (A) Domain of  $f(x)$  contains  $(-\infty, -1) \cup (1, \infty)$
- (B) Range of  $f(x)$  contains  $\left[ 0, \frac{1}{e-1} \right]$
- (C) Domain of  $f(x)$  is  $\left( -\infty, \frac{e}{1-e} \right] \cup \left[ \frac{1}{e-1}, \infty \right)$
- (D) Range of  $f(x)$  is  $(-\infty, -1] \cup (1, \infty)$
5. Let  $f(x)$  be a non constant twice derivable function defined on  $\mathbb{R}$  such that  $f(2+x) = f(2-x)$  and  $f' \left( \frac{1}{2} \right) = 0 = f'(1)$ , then minimum number of roots of the equation  $f''(x) = 0$  in  $(0, 4)$ , is
- (A) 4 (B) 3 (C) 2 (D) 5

Space for Rough Work

6. Let  $F(n, x) = \sum_{r=1}^n \tan^{-1} \left( \frac{2x}{4 + r(r+1)x^2} \right)$ , where  $x > 0$  and  $n \in \mathbb{N}$  and  $\frac{d}{dx} F(n, x) = G(n, x)$ , then
- (A)  $F(8, 2) = \tan^{-1} \left( \frac{4}{5} \right)$  (B)  $\lim_{n \rightarrow \infty} F(n, 2\sqrt{3}) = \frac{\pi}{6}$
- (C) If  $G\left(n, \frac{1}{3}\right) < 0$ , then  $n \geq 36$  (D)  $\lim_{n \rightarrow \infty} G(n, 1) = -\frac{2}{5}$
7. If  $a + b + c = 4$  and  $a^2 + b^2 + c^2 + 3(ab + bc + ac) = 21$  where  $a, b, c \in \mathbb{R}$ , then which of the following is/are true
- (A) minimum value of  $(a+b)(b+c)(a+c)$  is 18
- (B) minimum value of  $(a+b)(b+c)(a+c)$  is 24
- (C) maximum value of  $(a+b)(b+c)(a+c)$  is  $\frac{490}{27}$
- (D) maximum value of  $(a+b)(b+c)(a+c)$  is  $\frac{471}{27}$
8. Let  $f(x) = \begin{cases} x[x] & -4 \leq x < -1 \\ 1-x & -1 \leq x < 1 \\ x^2 - 3x + 2 & 1 \leq x < 2 \\ (x-2)[x] & 2 \leq x \leq 4 \end{cases}$ , then (where  $[.]$  denotes greatest integer function)
- (A) number of points where  $f(x)$  is non-differentiable is 6
- (B) sum of global maximum value and global minimum value of  $f(x)$  is  $\frac{63}{4}$
- (C) number of points of extrema of  $f(x)$  is 3 points
- (D) number of solutions of  $f(x) = 4$  is 3

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:  
**Full Marks** : +4 If only (all) the correct option(s) is (are) chosen.  
**Partial Marks** : +3 If all the four options are correct but **ONLY** three options are chosen.  
**Partial Marks** : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.  
**Partial Marks** : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.  
**Zero Marks** : 0 If none of the options is chosen (i.e. the question is unanswered).  
**Negative Marks** : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

Paragraph for Questions 9 and 10

$$\text{Let } \int_0^x g(t) dt = 3x + \int_x^0 (\cos^2 t) g(t) dt \text{ for } x > 0 \text{ and } f(x) = \lim_{\alpha \rightarrow 0} \frac{12}{\alpha^4} \int_0^\alpha \frac{(e^{(x+t)} - e^x) \ln^2(1+t)}{(2t^3 + 3)} dt$$

9. Which option(s) is/are correct -

(A) Maximum value of  $g(x)$  is 3

(B) Minimum value of  $g(x)$  is  $\frac{3}{2}$

(C)  $\int_0^\infty g(x) dx = \frac{3\pi}{2\sqrt{2}}$

(D)  $\int_0^\infty g(x) dx = \frac{3\pi}{4\sqrt{2}}$

10. Which option(s) is/are correct -

(A)  $\int f(x) (\sin x + \cos x) dx = \sin x f(x) + C$  (where  $C$  is the constant of integration)

(B)  $\int f(x) (\sin x - \cos x) dx = \sin x f(x) + C$  (where  $C$  is the constant of integration)

(C) Number of solutions of  $f(x) + |x| = 4$  is 2

(D) Number of solutions of  $f(x) + |x| = 4$  is 1

Space for Rough Work

**Paragraph for Questions 11 and 12**

Consider a function  $f(x) = ax^3 + x^2 - (2a + 1)x + 2$  (where  $a \geq 0$ )

11. If the number of value(s) of 'a' for which Rolle's theorem can be applied in the interval  $[0, 1]$  is equal to k, then k is less than or equal to
- (A) 0 (B) 1  
(C) 2 (D) 4
12. If  $k_1$  and  $k_2$  ( $k_1 \neq k_2$ ) are respectively the local maximum and local minimum values of  $f(x)$ , then the sum of those roots of the equations  $f(x) = k_1$  and  $f(x) = k_2$  at which  $f'(x) \neq 0$ , is
- (A)  $\frac{-2}{3a}$  (B)  $\frac{-1}{a}$   
(C)  $\frac{-2}{a}$  (D)  $\frac{-1}{3a}$

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**Space for Rough Work**

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +3 If ONLY the correct numerical value is entered.

*Zero Marks* : 0 In all other cases.

1. If  $\int \frac{x^{17}}{(1+x^2)^{10}} dx = \frac{1}{n} \left( \frac{x^2}{1+x^2} \right)^m + C$  where  $m, n \in \mathbb{N}$  and  $C$  is constant of integration, then value of  $\frac{m}{n}$  (in its lowest form) is equal to
2. A circle of radius  $2\sqrt{10}$  cm is divided by chord of length 12 cm into two parts. If maximum area of rectangle inscribed into smaller circular segment is  $\lambda$ , then  $\frac{\lambda}{\sqrt{15}}$  is
3. If the range of the function  $y = |f(x)|$ , where  $f(x) = \frac{195x(x^2+4)}{9x^4+97x^2+144}$  is  $\left[0, \frac{a}{b}\right]$  where  $a$  and  $b$  are relatively prime natural numbers then the value of  $a - 4b$  is
4. If  $\int \frac{dx}{(\sec x + \cos \sec x + \tan x + \cot x)^2} = \frac{x}{\alpha} + \frac{\cos(x + \frac{\pi}{4})}{\beta} + \frac{\cos 2x}{\gamma} + \delta$ , then  $|\alpha + \beta + \gamma|$  is
5. If  $f(x) = \max\{\min(x, x^2), \max(x, x^2)\}$ , where  $x \geq 0$  then  $\frac{d}{dx} \left( (f \circ f)^{-1}(x) \right)$  at  $x = 16$  is
6. Let  $f(x) = x^3 - 3x^2 + 3x + 1$  and  $g$  be inverse of it, then area bounded by the curve  $y = g(x)$  with  $x$  axis between  $x = 1$  and  $x = 2$ , is (in square units)
7. The value of the integral  $I = \int_1^3 \left( \tan^{-1} \left( \frac{x^2 - 5x + 6}{x^3 - 6x^2 + 12x - 7} \right) + \tan^{-1} \left( \frac{1}{x^2 - 4x + 4} \right) \right) dx$  is
8. If  $d = (x_2 - x_1)^2 + (-x_2 - 4 + 2\sqrt{x_1 + 2})^2$  where  $x_1 > -2, x_2 \in \mathbb{R}$ , then the minimum value of  $d$  is equal to -

Space for Rough Work

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Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. ....

I have read all the instructions  
and shall abide by them.

\_\_\_\_\_  
Signature of the Candidate

I have verified the identity, name and Form  
number of the candidate, and that question  
paper and ORS codes are the same.

\_\_\_\_\_  
Signature of the Invigilator

Space for Rough Work